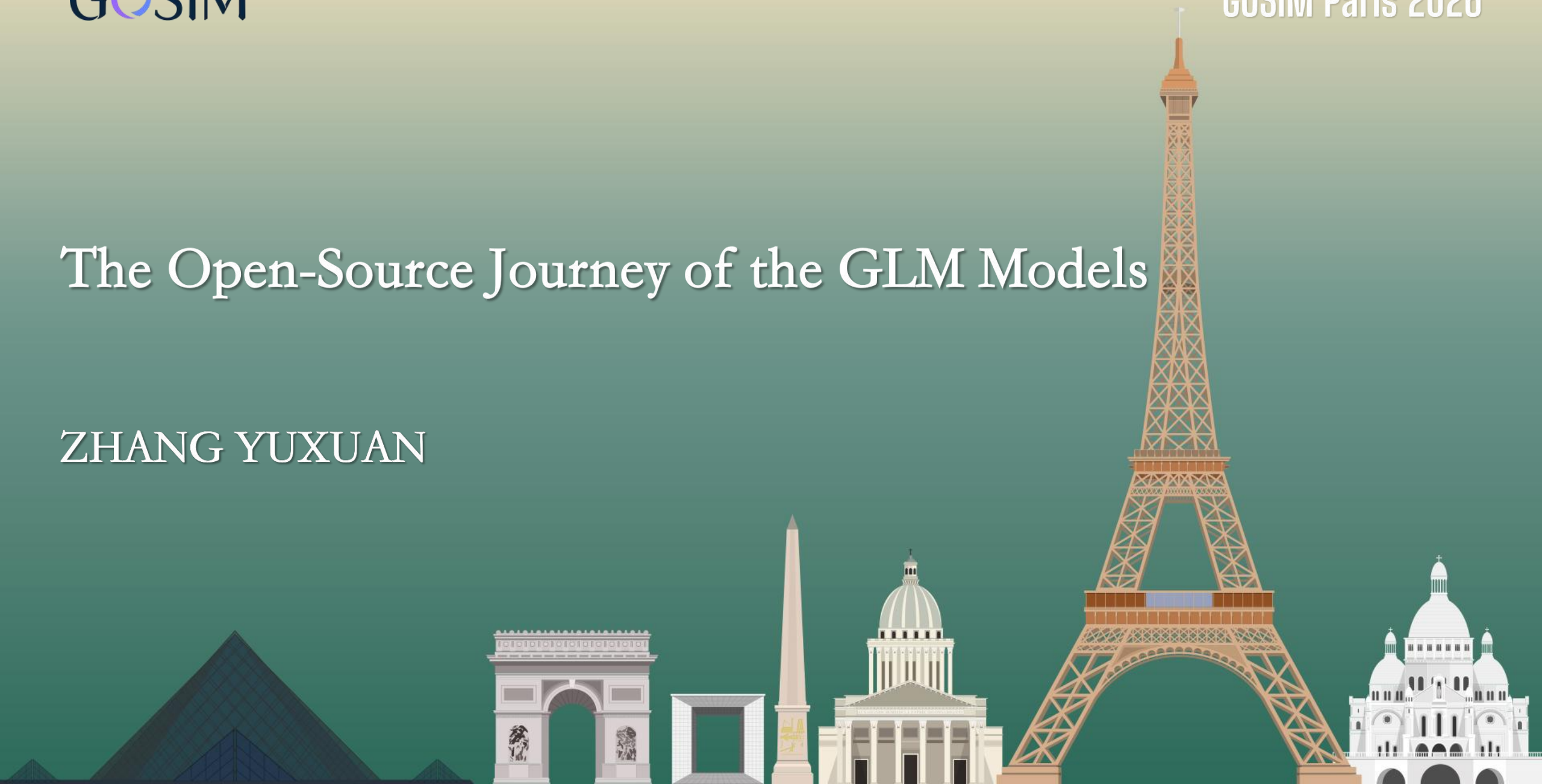


The Open-Source Journey of the GLM Models

ZHANG YUXUAN



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GOSIM Paris 2026

Part 01

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Part 03

Try GLM



01

GLM Series Model



Zhipu AI (Z.AI) was established through the commercialization of Tsinghua University's technological achievements, combining the mission of AGI with a commitment to domestic large language models. It has consistently aimed to benchmark against international frontiers and serve as a pioneer in Chinese LLM technological breakthroughs, maintaining an innovative spirit of "from 0 to 1" exploration in cutting-edge technologies.

- China's first and among the earliest globally to develop AGI large language models
- Since its establishment, has successively launched China's first ten-billion parameter model, first open-source hundred-billion parameter model, first conversational model, first multimodal model, world's first device control agent, etc.
- Continuously explores the boundaries of AGI technology and practical applications, with each release aimed at driving breakthrough technological innovation and producing disruptive results in Chinese LLMs

Top 3 globally, #1 in China, #1 in open-weight: Latest foundation model **GLM-5**
World's first deep research and agentic operation agent: AutoGLM Deep Think

First Chinese LLM to enter SuperBench top 3:
Foundation model **GLM-4-Plus**
World's first device control agent: AutoGLM

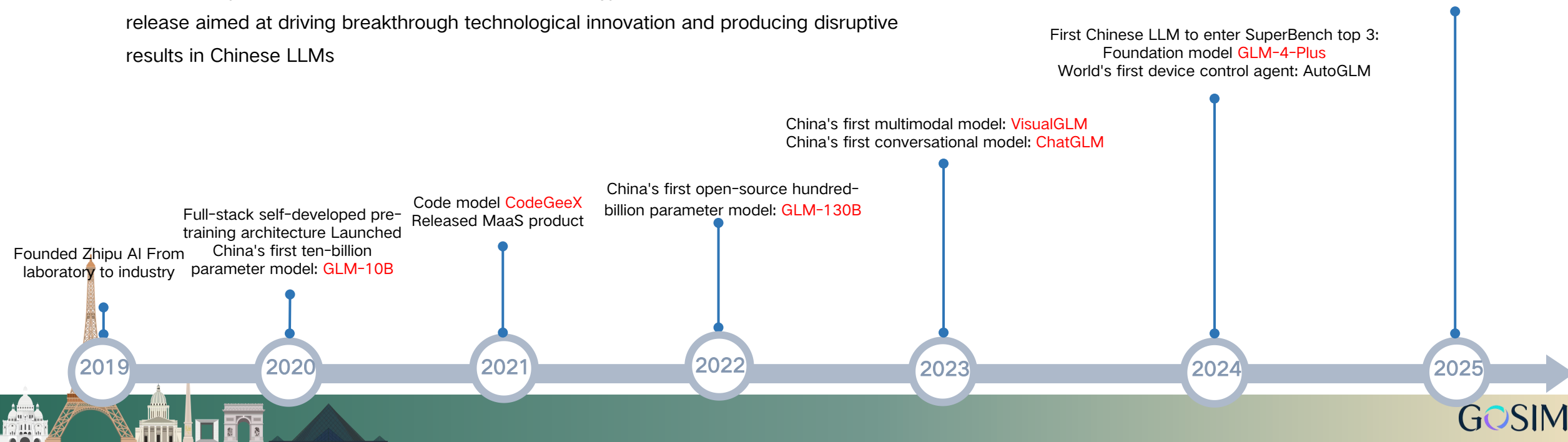
China's first multimodal model: **VisualGLM**
China's first conversational model: **ChatGLM**

China's first open-source hundred-billion parameter model: **GLM-130B**

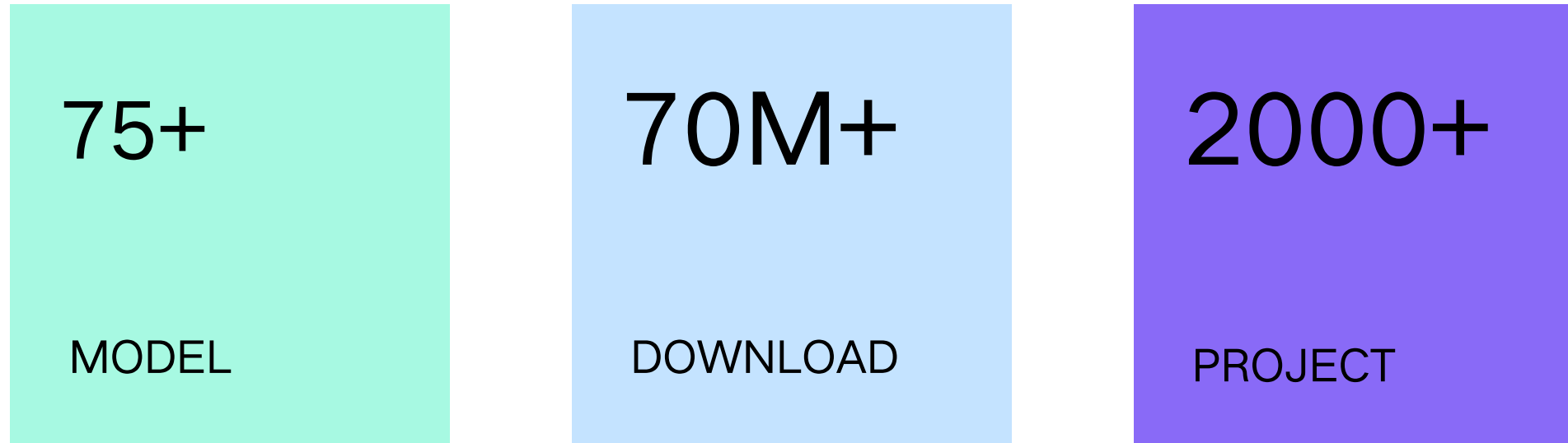
Code model **CodeGeeX**
Released MaaS product

Full-stack self-developed pre-training architecture Launched
China's first ten-billion parameter model: **GLM-10B**

Founded Zhipu AI From laboratory to industry







A rough and conservative count of secondary development projects found on GitHub by searching keywords like “glm”



**WHAT IF
A MODEL
CAN CODE FOR
8 HOURS
STRAIGHT?**



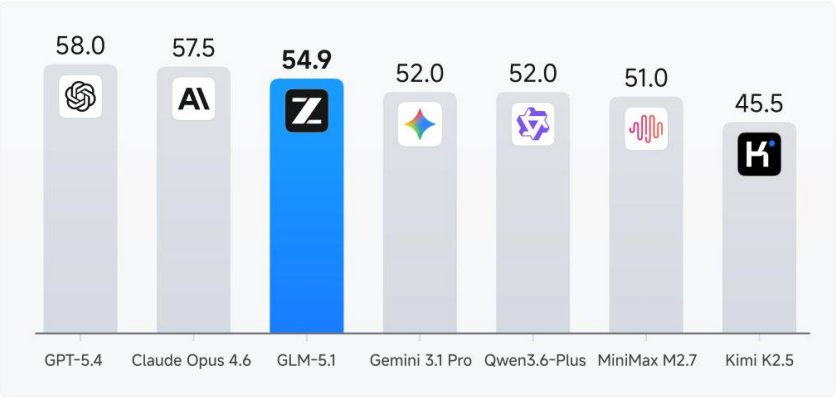
Core Advantages & Technical Highlights

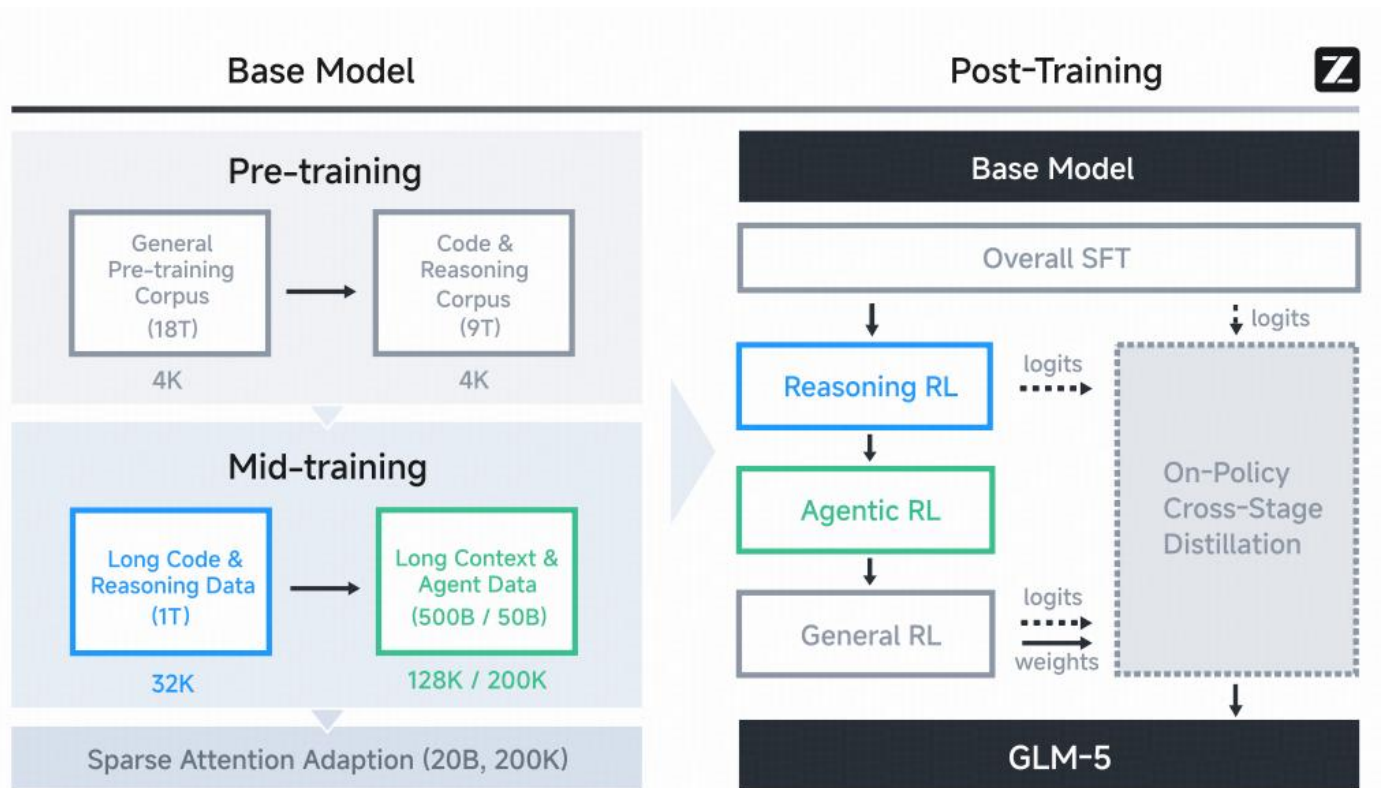
- 8-Hour Long-Horizon Tasks
- Full loop of autonomous planning → execution → self-evolution
- Strong Performance
- First open-source model fully aligned with Claude Opus 4.6
- 200K Context + MCP
- Native tool use, multi-step reasoning, Artifacts front-end generation
- Coding Plan Lite/Pro/Max
 - 8h Autonomous METR leaderboard
 - MIT Open Source HuggingFace weights, commercial/fine-tune
 - MCP Integration external tools, multi-step

Benchmark	GLM-5.1	GLM-5	Qwen3.6-Plus	MiniMax M2.7	DeepSeek-V3.2	Kimi K2.5	Claude Opus 4.6	Gemini 3.1 Pro	GPT-5.4
REASONING									
HLE	31.0	30.5	28.8	28.0	25.1	31.5	36.7	45.0	39.8
HLE w/ Tools	52.3	50.4	50.6	-	40.8	51.8	53.1*	51.4*	52.1*
AIME 2026	95.3	95.4	95.1	89.8	95.1	94.5	95.6	98.2	98.7
HMMT Nov. 2025	94.0	96.9	94.6	81.0	90.2	91.1	96.3	94.8	95.8
HMMT Feb. 2026	82.6	82.8	87.8	72.7	79.9	81.3	84.3	87.3	91.8
IMOAnswerBench	83.8	82.5	83.8	66.3	78.3	81.8	75.3	81.0	91.4
GPQA-Diamond	86.2	86.0	90.4	87.0	82.4	87.6	91.3	94.3	92.0
CODING									
SWE-Bench Pro	58.4	55.1	56.6	56.2	-	53.8	57.3	54.2	57.7
NL2Repo	42.7	35.9	37.9	39.8	-	32.0	49.8	33.4	41.3
Terminal-Bench 2.0 Terminus-2	63.5	56.2	61.6	-	39.3	50.8	65.4	68.5	-
Terminal-Bench 2.0 Best self-reported harness	69.0 (Claude Code)	56.2 (Claude Code)	-	57.0 (Claude Code)	46.4 (Claude Code)	-	-	-	75.1 (Codex)
CyberGym	68.7	48.3	-	-	17.3	41.3	66.6	38.8	66.3
AGENTIC									
BrowseComp	68.0	62.0	-	-	51.4	60.6	-	-	-
BrowseComp w/ Context Manage	79.3	75.9	-	-	67.6	74.9	84.0	85.9	82.7
τ³-Bench	70.6	69.2	70.7	67.6	69.2	66.0	72.4	67.1	72.9
MCP-Atlas Public Set	71.8	69.2	74.1	48.8	62.2	63.8	73.8	69.2	67.2
Tool-Decathlon	40.7	38.0	39.8	46.3	35.2	27.8	47.2	48.8	54.6
Vending Bench 2	\$5,634.41	\$4,432.12	\$5,114.87	-	\$1,034.00	\$1,198.46	\$8,017.59	\$911.21	\$6,144.18

Coding Performance Evaluation

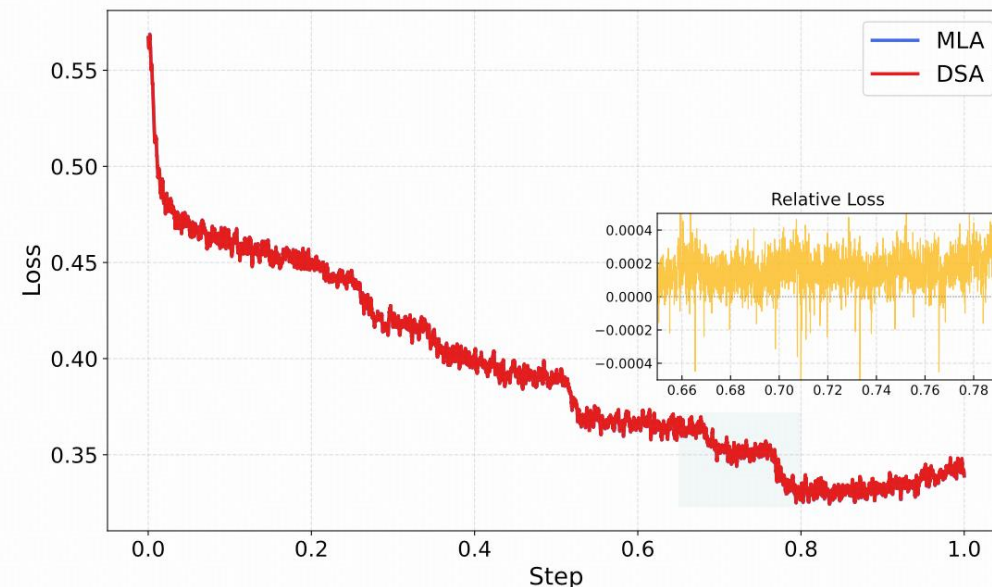
3 Benchmarks: SWE-Bench Pro, Terminal-Bench 2.0, NL2Repo

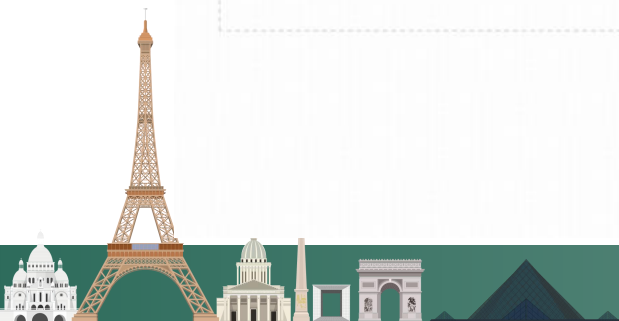
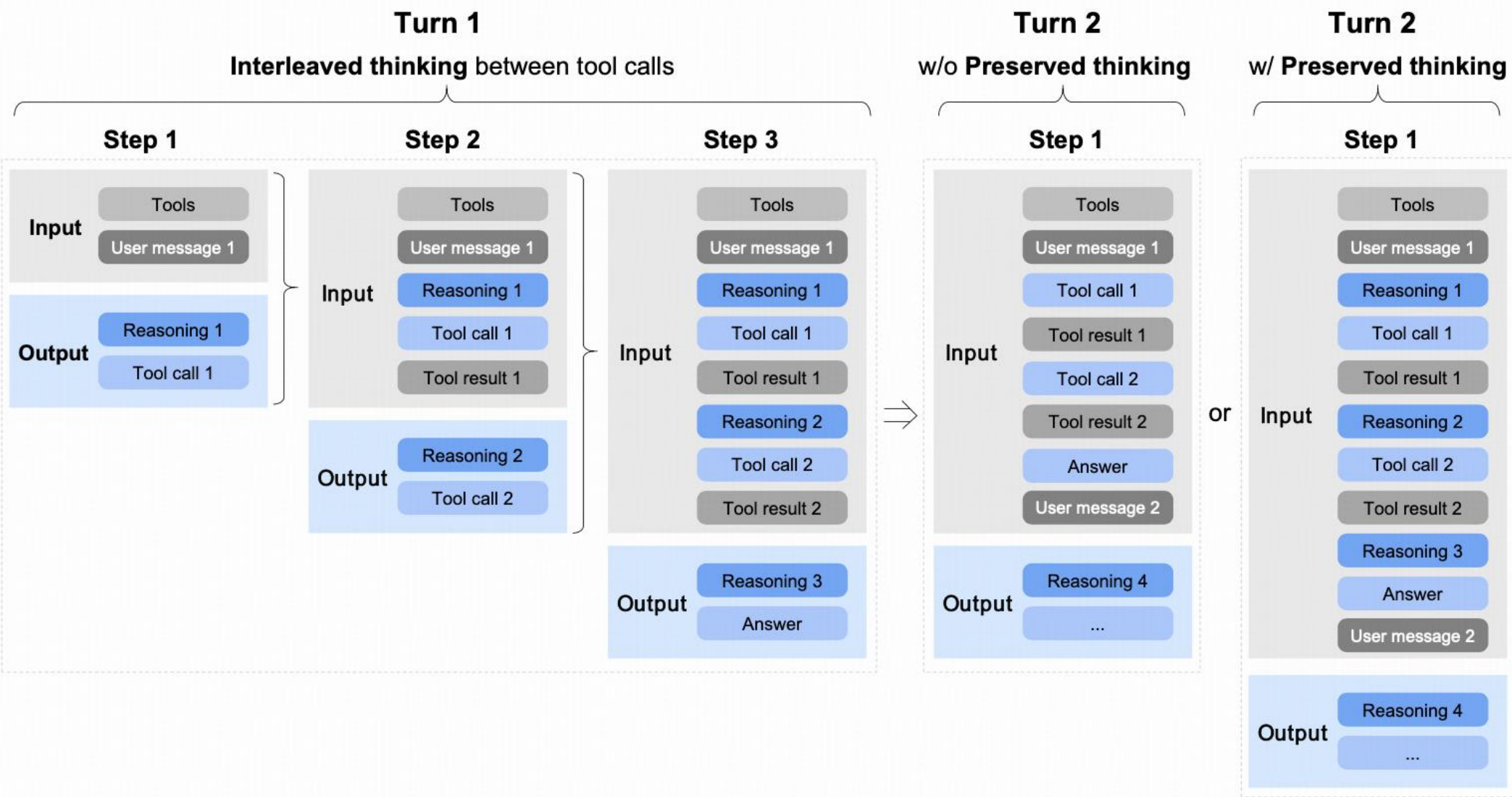


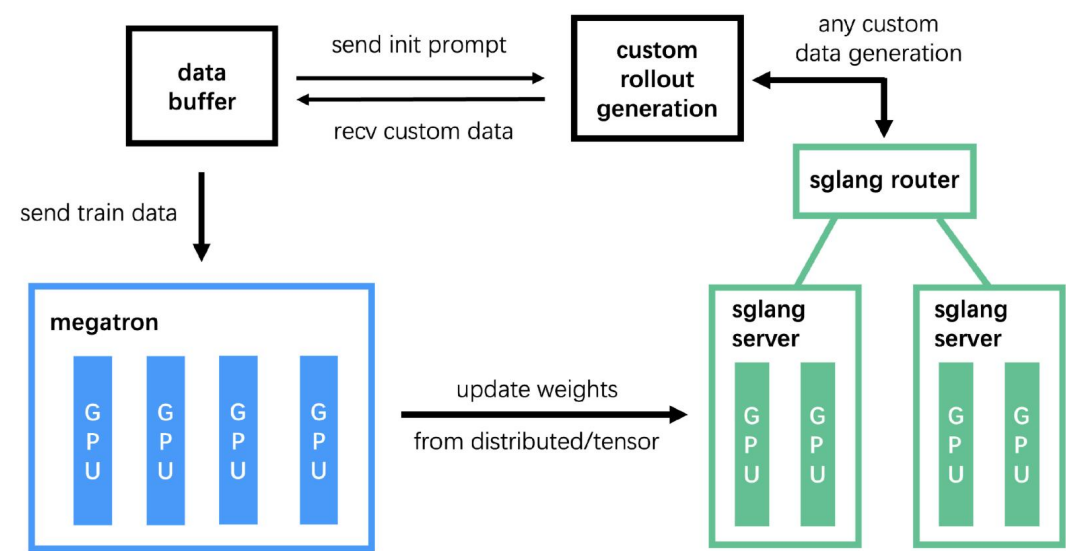


Training Pipeline

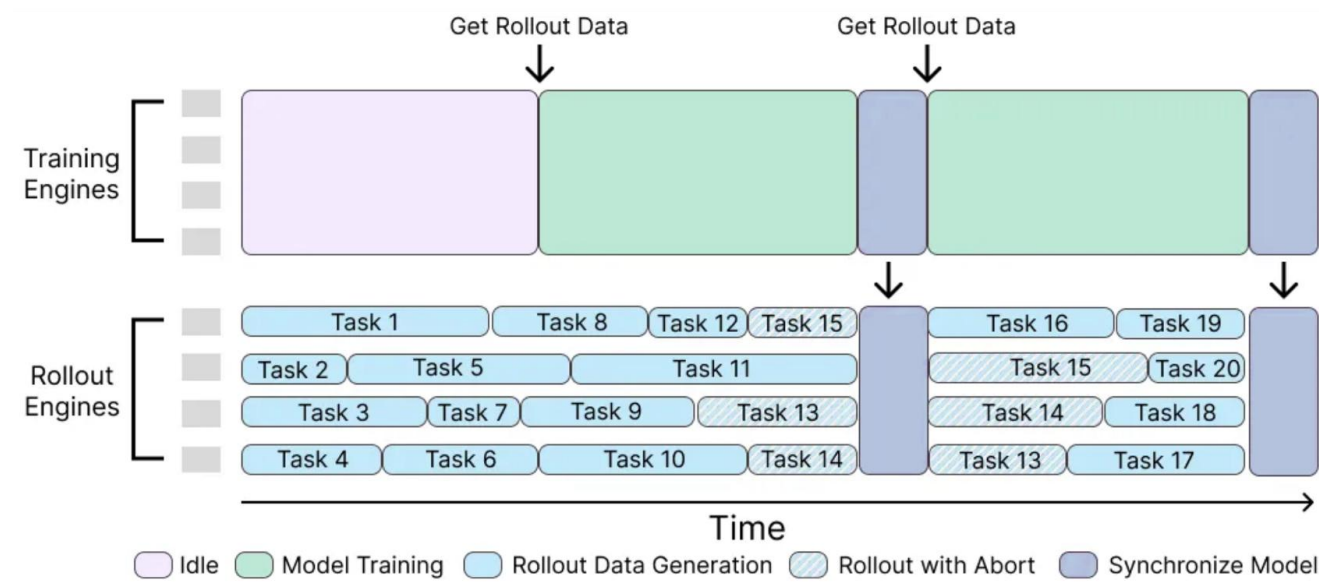
SFT Loss Comparison: DSA vs. MLA





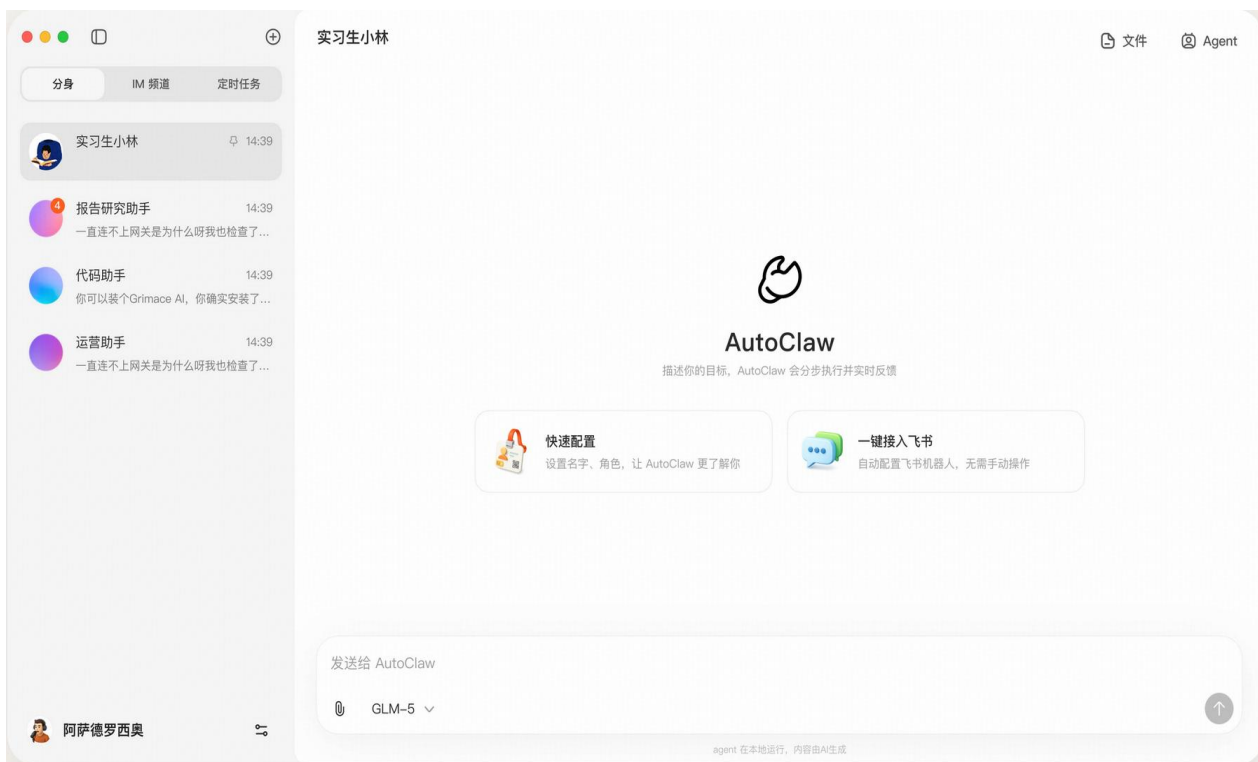


Asynchronized and decoupled training and inference



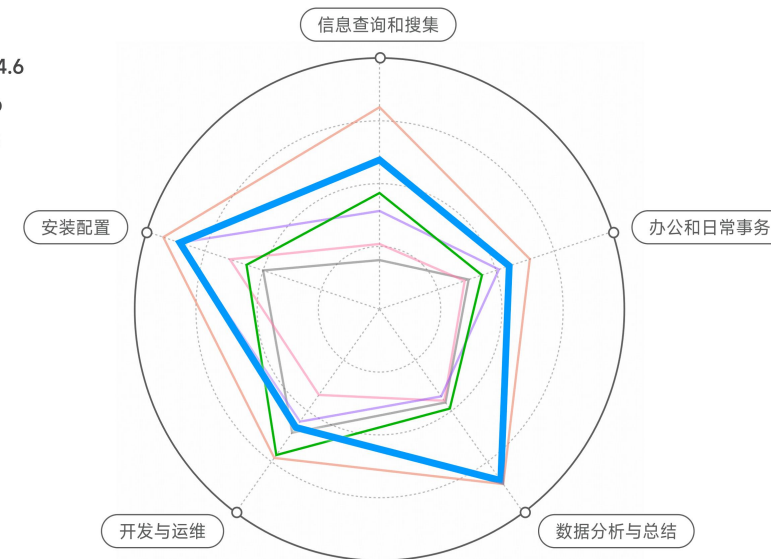
Support large-scale and multi-task agent training





ZClawBench: 龙虾场景能力评测

- GLM-5-Turbo
- GLM-5
- Claude Opus 4.6
- Gemini 3.1 Pro
- MiniMax M2.5
- Kimi K2.5



GLM-5-Turbo is a foundation model deeply optimized for the OpenClaw scenario. It has been specifically optimized for the core requirements of OpenClaw tasks since the training phase, enhancing key capabilities such as tool invocation, command following, timed and persistent tasks, and long-chain execution.

<https://huggingface.co/datasets/zai-org/ZClawBench>



GLM-5V-Turbo

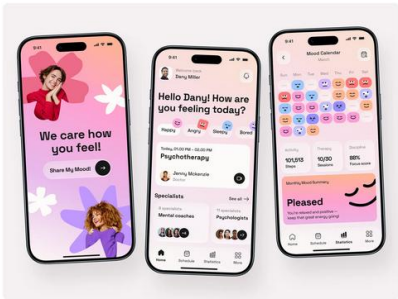
Benchmarks

		GLM-5V-Turbo	Kimi K2.5	Claude Opus 4.6
Multimodal Coding	Design2Code	94.8	91.3	77.3
	Flame-VLM-Code	93.8	88.8	98.8
	Vision2Web	31.0	33.2	43.5
Multimodal ToolUse	ImageMining	30.7	24.4	-
	BrowseComp-VL	51.9	42.9	35.9
	MMSearch	72.9	58.7	63.8
	MMSearch-Plus	30.0	25.6	25.6
	SimpleVQA	78.2	71.5	63.2
	Facts	58.6	57.8	-
	V*	89.0	84.3	66.5
GUI Agent	OSWorld	62.3	63.3	72.2
	AndroidWorld	75.7	43.1	62.0
	WebVoyager	88.5	84.3	88.0

- Built for Multimodal Coding
- Native Multimodal Inputs
- Designed for Long-Horizon Tasks
- Ready for Agentic Workflows

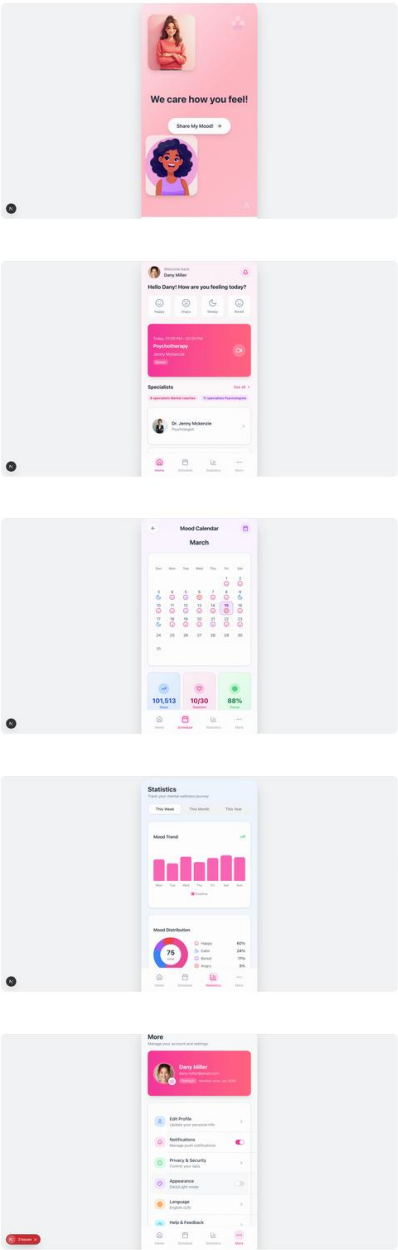


Input



Please recreate the mobile pages based on the design mockups in the images. The left side shows the welcome page, and the center shows the homepage image. You will also need to create mockups for the remaining two pages.

Output



GLM-Image-Generation

Official skill for generating high-quality images from text prompts using ZhiPu GLM-...

★ 0 · 📦 3

GLM-OCR-Table

Recognize and extract tables from images and PDFs into Markdown format using ZhiPu...

★ 0 · 📦 18

GLM-OCR-Handwriting

Recognize handwritten text from images using ZhiPu GLM-OCR API. Supports various...

★ 0 · 📦 19

GLM-OCR-Formula

Recognize and extract mathematical formulas from images and PDFs into LaTeX...

★ 0 · 📦 21

GLM-V-Resume-Screen

Screen and evaluate resumes against criteria using ZhiPu GLM-V multimodal model. Rea...

★ 0 · 📦 22

GLM-V-Prompt-Gen

Analyze images/videos and generate professional prompts for text-to-image and text-to-...

★ 0 · 📦 19

GLM-V-Caption

Generate captions (descriptions) for images, videos, and documents using...

★ 0 · 📦 18

GLM-OCR

Extract text from images using GLM-OCR API. Supports images and PDFs with high accuracy...

★ 1 · 📦 32

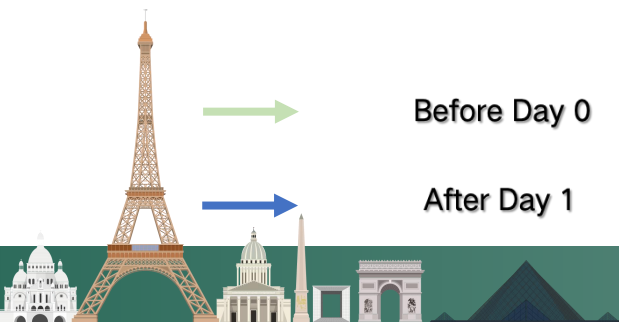
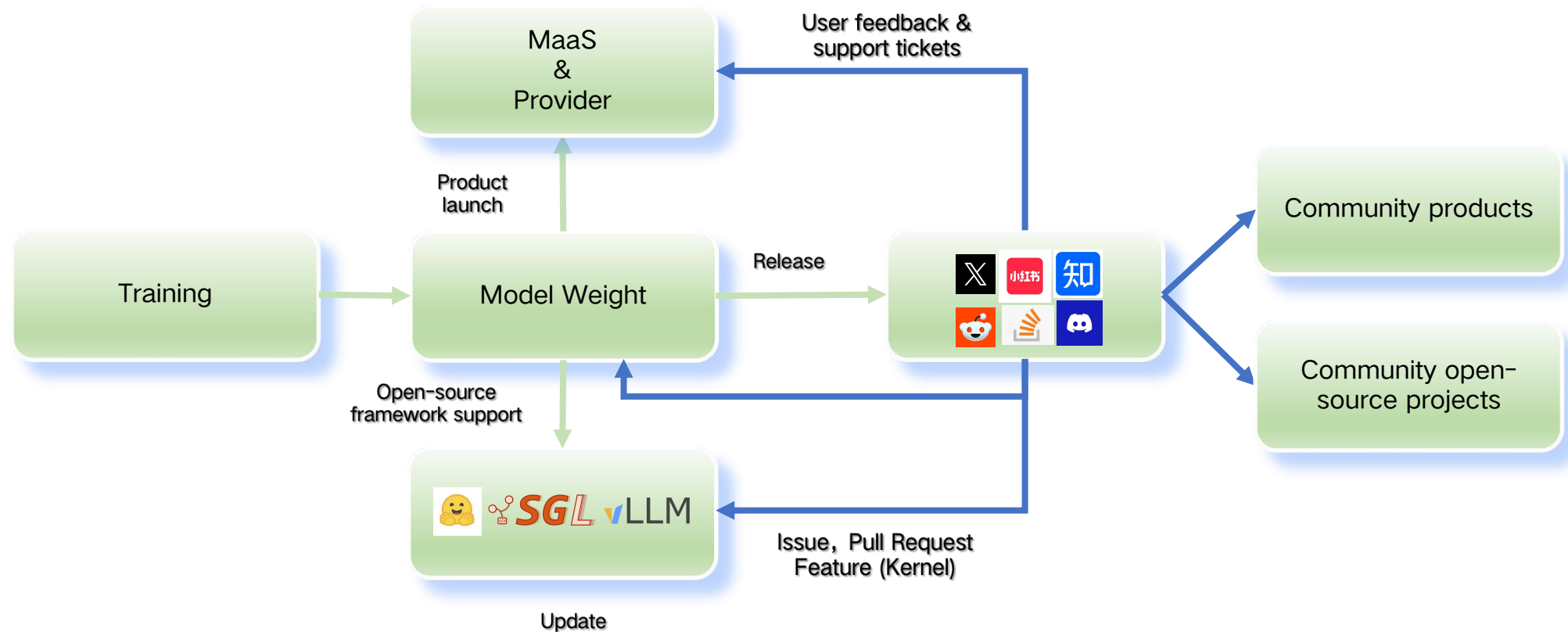
The **GLM-OCR / GLM-V** skill series continues to roll out, with targeted optimizations for pain-point scenarios such as **handwriting recognition, table recognition, and formula recognition**.

The **GLM-OCR technical report** has also been released, and the **pip package** has been updated. For best results, it is recommended to use it together with the **skills**.



02 GLM OSS Ecosystem





- Open-sourced upon release.
- Actively welcome feedback and address issues.

FP4 / W8A8

NPU / XPU
Hardware
support

Relevant issues: >300 (~20% remaining)

VIT DP / TP

PD
disaggregation

Relevant PRs: >200 (~10% remaining)

Kernel

KV Cache

> These statistics include data from SGLang, vLLM,
and the official GLM repositories.



2026-04-30 · Research

ABNORMAL OUTPUT RATE

0.1% → <0.03%*after fixing the KV Cache race in PD-disaggregation*

THROUGHPUT GAIN

+10% to +132%*LayerSplit on 40K–120K token contexts*

FAILURE FREQUENCY

3–5 / 10,000*anomalies surfaced only at peak production load*<https://github.com/sgl-project/sglang/pull/22811>

Scaling Pain of Coding Agent Serving: Lessons from Debugging GLM-5 at Scale

Our belief in Scaling Laws has not only driven continuous breakthroughs in model parameters and data scale, but has also pushed infrastructure engineering toward its limits. This process inevitably comes with growing pains, which we refer to as **Scaling Pain**.

As large language model applications move beyond simple dialogue and toward more complex, long-horizon Coding Agent tasks, our inference infrastructure has come under unprecedented pressure, serving hundreds of millions of Coding Agent requests every day. Over the past few weeks, some users encountered several types of abnormal outputs when using the GLM-5 series for complex Coding Agent tasks, including garbled output, repetition, and rare-character generation.

These issues do not appear under standard inference settings. They only emerge in high-concurrency, long-context Coding Agent workloads, making them extremely difficult to reproduce reliably. After several weeks of investigating, debugging, and stress testing, the team eventually identified and fixed several independent low-level race-condition bugs. We also implemented targeted optimizations for the underlying system bottlenecks revealed during this process, significantly improving both the stability and efficiency of the inference system.

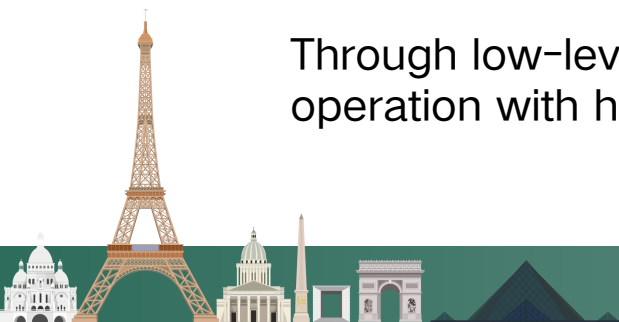
In this post, we share the lessons learned from this investigation, with the hope of helping the community better understand and overcome the Scaling Pain of Coding Agent inference.





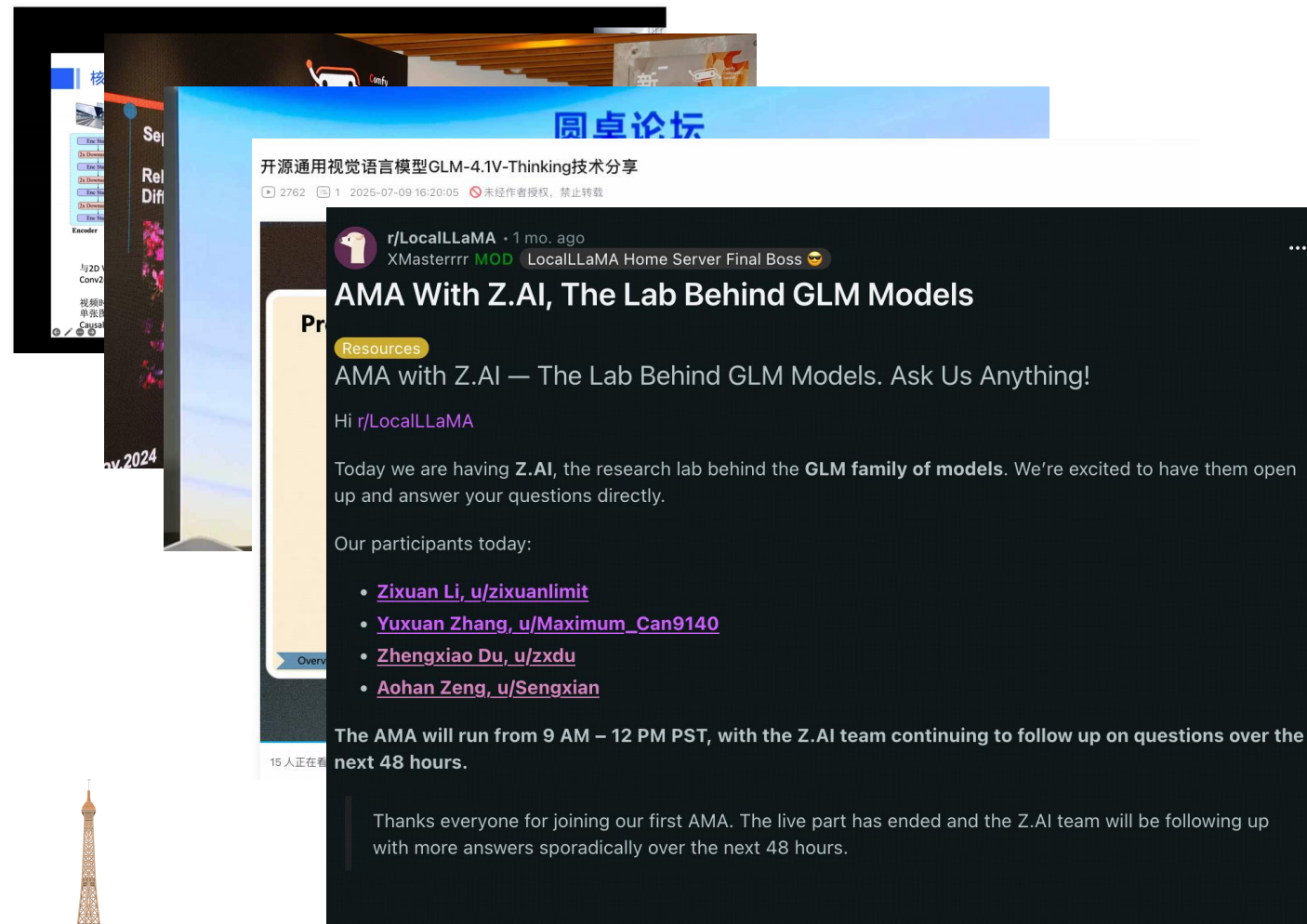
GLM-5 has completed deep inference adaptation with major domestic computing platforms including Huawei Ascend, Moore Threads, Cambricon, KUNLUNXIN, MetaX, Enflame, and Hygon.

Through low-level operator optimization and hardware acceleration, GLM-5 has achieved stable operation with high throughput and low latency on Chinese chip clusters



4090单卡可跑 | CogVideoX视频生成模型

4817 0 2024-08-09 16:05:11 未经作者授权，禁止转载



we have actively participated in and organized technical briefings and open-source ecosystem events — the main ways we promote the model domestically.

03

Try GLM



A stylized illustration of Parisian landmarks. The Eiffel Tower is the central focus, rendered in a golden-brown color. To its left is the white dome of the Louvre Museum. To its right is the white dome of the U.S. Capitol building. Further right is the Washington Monument, a tall, thin, white obelisk. Next to it is the Arc de Triomphe, a white archway. On the far right is the Great Pyramid of Giza, a large, dark, triangular structure. The background is a solid dark green, and the sky is a light blue gradient.



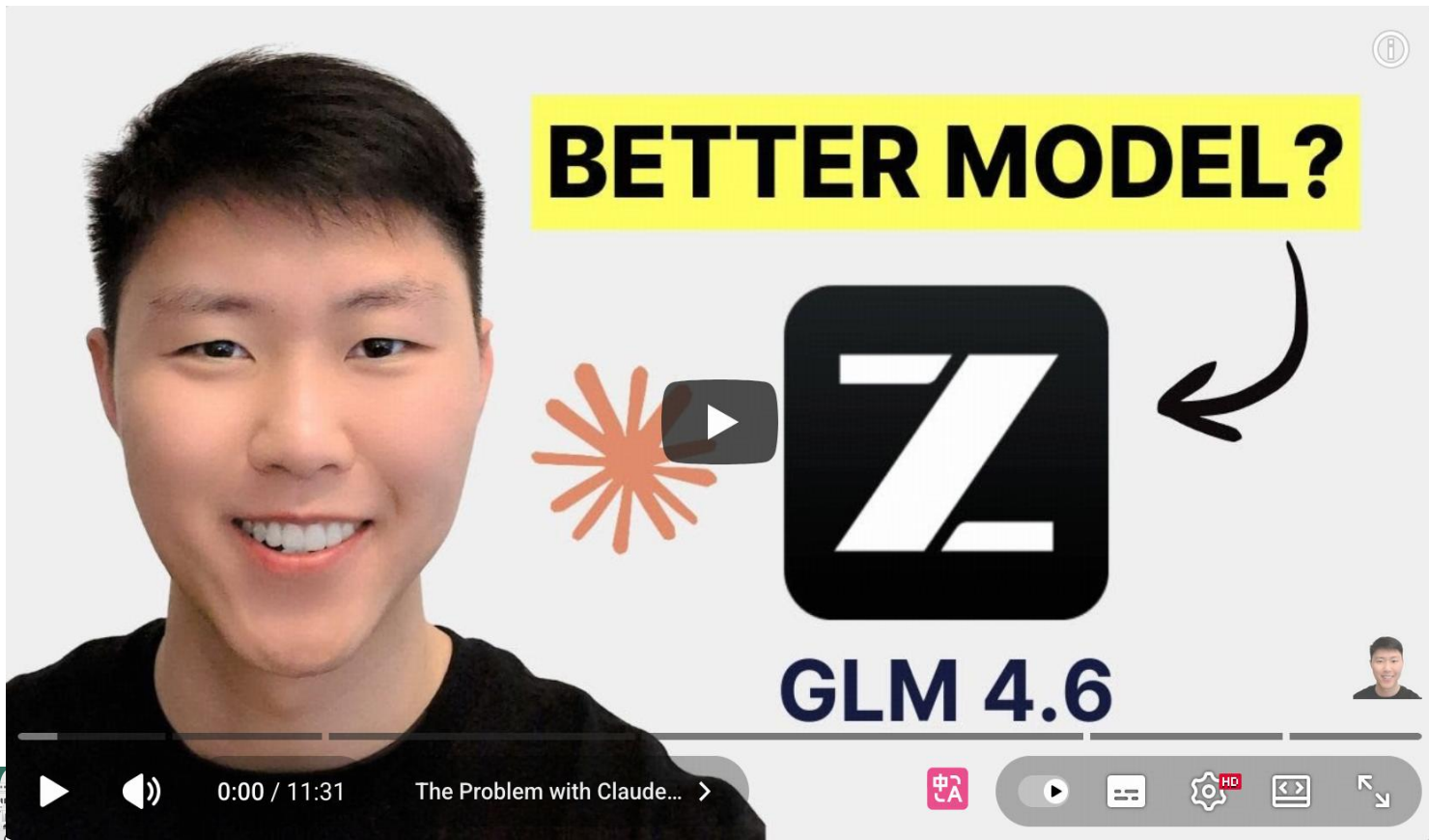
GLM Coding Plan

If you haven't tried GLM Coding Plan, it is highly recommended for your coding productivity:

<https://z.ai/subscribe>

For GLM Coding Plan Setup and Test, you can watch the setup and demonstration video:

<https://www.youtube.com/watch?v=P6QIH0F06L8&t=163s>



[Coding Plan](#)

[Released Notes](#)

[Terms and Policy](#)

[Help Center](#)

Campaign Rules

Invite Friends, Get Credits

[Copy page](#)



Please read and fully understand all the rules below before participating in the campaign. Participation constitutes your acceptance and agreement to be bound by these rules.

Last Update: September 18, 2025

1. General Principles

- **Campaign Organizer:** This campaign is initiated and operated by the service provider of Z.ai Platform (hereinafter referred to as "we," "us," or the "Platform").
- **Campaign Purpose:** To reward users and enhance product experience, we are launching the "Invite Friends, Get Credits" campaign.
- **Campaign Nature:** This campaign offers incentive Credits. All Credits are preferential benefits provided by the Platform, **not cash, currency, or financial instruments**, and do not possess monetary attributes.
- **User Agreement:** By participating, you represent that you have **read, understood, and agreed** to be bound by these rules. We reserve the right to modify or adjust these rules based on laws, regulations, and operational needs. Changes will be announced on the campaign page in advance.

Access on API and Product

- API: <https://docs.bigmodel.cn/cn/api/introduction>
- AutoClaw: <https://autoglm.zhipuai.cn/autoclaw>

Tech Report

- GLM-5.1: <https://z.ai/blog/glm-5.1>
- GLM-5: <https://arxiv.org/abs/2602.15763>
- GLM-OCR: <https://arxiv.org/abs/2603.10910>
- GLM-5V-Turbo: <https://arxiv.org/pdf/2604.26752>



Thanks!

